

Golden Ratio Tight-Binding Chain: Finite-Size Scaling

QSOL LLC — Aug 16 2026 Termux exact diagonalization SHA3-512 + JCS

Repo: github.com/qsolllc/golden-ratio-tb anchor 42e34f88fca79abf genesis 1e93f575d54cc906d577164c75ffcf1c5fa7512d

Fibonacci finite-size scaling <IPR>

L	lam0.5	lam2.0	lam2.5
13	0.1248	0.3406	0.5283
21	0.0804	0.2675	0.4703
34	0.0508	0.2234	0.4498
55	0.0323	0.1746	0.4418

Extended $\sim 1/L$ collapse, localized $\sim \text{const}$

Chain provenance

- STITCH-0000 genesis_hash aa94d42b...
- POCATELLO 46/45 stitches E0=0.15, final anchor 42e34f88

L=89 + Entanglement (Aug 16)

Scaling to L=89

- L=55 lam0.5 0.03232 -> L=89 0.02038 extended 1/L
- L=55 lam2.5 0.44176 -> L=89 0.43331 localized const

Entanglement spectrum L=21 lam0.5 ground state Nf=10

- cut7 S=0.8053 top eigenvalue 0.613
- cut10 S=0.7680 top eigenvalue 0.556

IPR: 0.1248->0.02038 extended 1/L vs 0.5283->0.43331 localized const

<r>: 0.7242 GOE -> 0.3529 Poisson 0.386 localized

Entanglement S=0.8053 cut7 top 0.613 S=0.7680 cut10 top 0.556

Fibonacci Pulse Engine: 8.03 12.98 21.01 33.99 55.01 ms = Φ^2 , coherence 0.21->0.99 LOCK